

# HACKATHON 2026

**OPENING DAY**



THE ENERGY TO INNOVATE

INTRODUCING

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# Mahashree Ajmeri

Master of Ceremonies



## WELCOME SPEECH

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# Christine Blake

Associate Vice President, Students





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# **WELCOME TO SENECA ENERGY HACKATHON 2026**

THANK YOU

## SPONSORS

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THANK YOU

**PARTNERS**

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# HACKATHON 2026 OVERVIEW



# SPRINT ROADMAP

## HACKATHON STRUCTURE

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**May 24–26**

### **Hacking Days**

Submission due on the 26th at 11:59 PM

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**May 27**

### **Qualifier Round Judging**

First-round judging determines top teams

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**May 28**

### **Qualifier Winners Announced**

Finalists revealed and invited

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**May 29**

### **Preparation of Final Presentations**

Finalist prep day

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**May 30**

### **Final Showcase & Winners**

Grand finale and awards

# VIRTUAL

## MENTORING SESSIONS – MAY 24

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**Daniel Noakes**

Technical Solutions Specialist, Esri Canada

**Domain Session**  
1:00 – 2:00 PM



**Jonathan Van Dusen**

Higher Education Specialist, Esri Canada

**Technical Session**  
2:00 – 3:00 PM



**Alex Smith, PhD**

Spatial Data Scientist, Esri Canada

**Technical Session**  
2:00 – 3:00 PM



**Kartik Sorathiya**

Founder of Octo

**Product Thinking & Innovation**  
3:00 – 4:00 PM

IN-PERSON | LOCATION: K2001

## MENTORING SESSIONS – MAY 25

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**Mark Buchner**

Professor & Hackathon Chair, Seneca  
Polytechnic

**Pitch Development**  
2:00 – 3:00 PM



**Jonathan Van Dusen**

Higher Education Specialist, Esri Canada

**Technical + Hands-on**  
2:00 – 3:00 PM



**Alex Smith, PhD**

Spatial Data Scientist, Esri Canada

**Technical + Hands-on**  
2:00 – 3:00 PM

VIRTUAL

## MENTORING SESSIONS – MAY 26

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*Recommended Pre-Read:* [Video Link | https://www.alectra.com/innovation-projects](https://www.alectra.com/innovation-projects)



**Keith Hemingway**

Head, Emerging Technology, GRE&T  
Centre, Alectra

**Domain Session**  
10:00 – 11:00 AM



**Daniel Carr**

Head, Grid Edge Solutions, GRE&T Centre,  
Alectra

**Domain Session**  
11:00 – 12:00 PM

## HACKATHON SUBJECT MATTER

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# Mark Buchner

Professor & Hackathon Chair



# WHAT WE'LL COVER

## AGENDA

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- 01** About Seneca Hackathon — history and features
- 02** The origin of the “Energy” theme
- 03** Challenge set discussion
- 04** Recommendations for your teams
- 05** Design thinking, research, student success, and mentoring
- 06** Guidelines and deliverables
- 07** Schedule, rubric, and tips

# ONE HACKATHON. TWO GROUPS. MANY IMPACTS.

## STUDENTS & PARTNERS

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### STUDENTS

- Experiential Learning — apply classroom knowledge to real-world energy challenges using industry data, GIS, and design thinking
- Resume Building — build a strong portfolio with projects, prototypes, and presentations
- Camaraderie, Teamwork & Networking — collaborate, learn, and build lasting connections
- Build Pathway to Innovation & Commercialization — turn ideas into viable solutions with real-world impact

### PARTNERS

- Talent Identification — engage with diverse student talent and identify future interns and employees early
- Innovation Harvesting — discover fresh ideas, creative approaches, and new perspectives to address real energy challenges
- Proof of Technology / Proof of Concept — evaluate early-stage solutions that can evolve into real-world projects and strategic collaborations

*Together, we accelerate innovation, build stronger communities, and power a sustainable energy future.*

## CONTEXT

### ONTARIO'S ENERGY SUPERPOWER VISION

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#### THE WORLD TODAY: HIGH UNCERTAINTY

- AI disruption accelerating
- US–Canada trade uncertainty
- Economic volatility & climate change intensifying
- Electrification driving unprecedented demand

*Energy infrastructure is becoming the foundation of economic competitiveness.*

#### WHY YOU MATTER; THIS IS YOUR MOMENT

- The province needs innovative ideas
- Smarter use of data and GIS
- AI and digital solutions
- Resilience and risk planning
- Community focused innovation
- Energy equity and inclusion
- Data driven decision making

#### ONTARIO'S OPPORTUNITY: A CLEAR VISSION

- Nuclear Expansion & Modernization
- Smart Grids & Grid Resilience
- Renewable Integration
- Electrification of Transportation & Buildings
- Energy Innovation & Clean Technology
- Community Benefits, Energy Equity & Climate Resilience

*Ontario's electricity demand is projected to rise 75% by 2050. — IESO, 2023 Forecast*

*Ontario's positioning itself as a clean energy superpower – Stephen Lecce, Ontario Minister of Energy and Electrification*



READ CAREFULLY

## FORMAL CHALLENGE SET DOCUMENTS

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Review the documents at [senecahackathon.com/challenges](https://senecahackathon.com/challenges)

- Review carefully with your team and select a particular problem statement.
- The problem statements are guidelines — you may improvise, though that's not advisable if you are new to the problem domain.

**Seneca**

### Hackathon 2026 Challenge Set Design – Final

#### What Students Can Build in Two Days

Students will have two days to produce either a simple prototype or a conceptual solution. Deliverables may include:

- Interactive dashboards
- GIS maps and spatial analyses
- Lightweight web or mobile prototypes
- Conceptual mockups or wireframes
- Data visualizations
- Predictive models or simulations
- Workflow diagrams or planning tools
- Story Maps or narrative presentations

Both technical and non-technical solutions are valid. Students are encouraged to focus on feasibility, clarity, and creative problem-solving rather than fully polished applications.

#### Theme 1: Clean Energy Generation and Integration

Canada's energy transition depends on integrating solar, wind, and distributed energy resources into existing grids. Utilities must identify where renewables can be added, how much capacity the grid can support, and how to maintain reliability as renewable adoption grows. This theme encourages students to explore renewable siting, DER planning, and microgrid opportunities using GIS and data-driven insights.

##### Problem Statement 1: Renewable Potential, Site Selection, and Load Considerations

**Problem:**

Cities and utilities need to understand where renewable (solar, wind, etc.) installations will be



# CHALLENGE THEMES

## ONE-PAGE CHALLENGE SET SUMMARY

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### Theme 1: Clean Energy Generation & Integration

*How can renewable energy be integrated into communities more effectively, intelligently, and sustainably?*

- Key Areas: Solar & wind suitability, renewable site selection, DERs, grid capacity planning, community microgrids, energy demand analysis

### Theme 2: Smart Grid Resilience & Electrification

*How can Ontario modernize and strengthen the electrical grid in response to climate change, electrification, and growing demand?*

- Key Areas: Weather-driven outages, climate resilience, infrastructure vulnerability, EV charging demand, smart grid modernization, predictive analytics

### Theme 3: Community Energy, Equity & Sustainability

*How can energy transition and climate resilience be made more equitable and inclusive for vulnerable communities?*

- Key Areas: Energy poverty, environmental justice, climate vulnerability, pollution exposure, community resilience, equity-focused planning

*Teams build GIS maps, dashboards, simulations, wireframes, and workflow diagrams. Both technical and non-technical solutions are welcome.*

# APPROACH

## DESIGN THINKING FOR ENERGY INNOVATION

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*Great solutions begin with understanding people, not technology.*

### 1. EMPATHIZE — Understand stakeholders and pain points

- **Who:** Residents, utilities, EV drivers, vulnerable communities, emergency services, businesses
- **Artifacts:** Interviews, stakeholder maps, personas

### 2. DEFINE → 3. IDEATE → 4. PROTOTYPE → 5. TEST

- **Define:** Clearly identify the real problem (e.g. grid overload risk, energy poverty, outage vulnerability)
- **Ideate:** Generate innovative solution concepts — GIS dashboards, AI forecasting, mobile apps, planning tools
- **Prototype:** Build lightweight proof-of-concepts — wireframes, GIS maps, dashboards, UML diagrams, story maps
- **Test:** Evaluate feasibility and impact — would stakeholders use this? Does it solve the problem? Is it realistic?

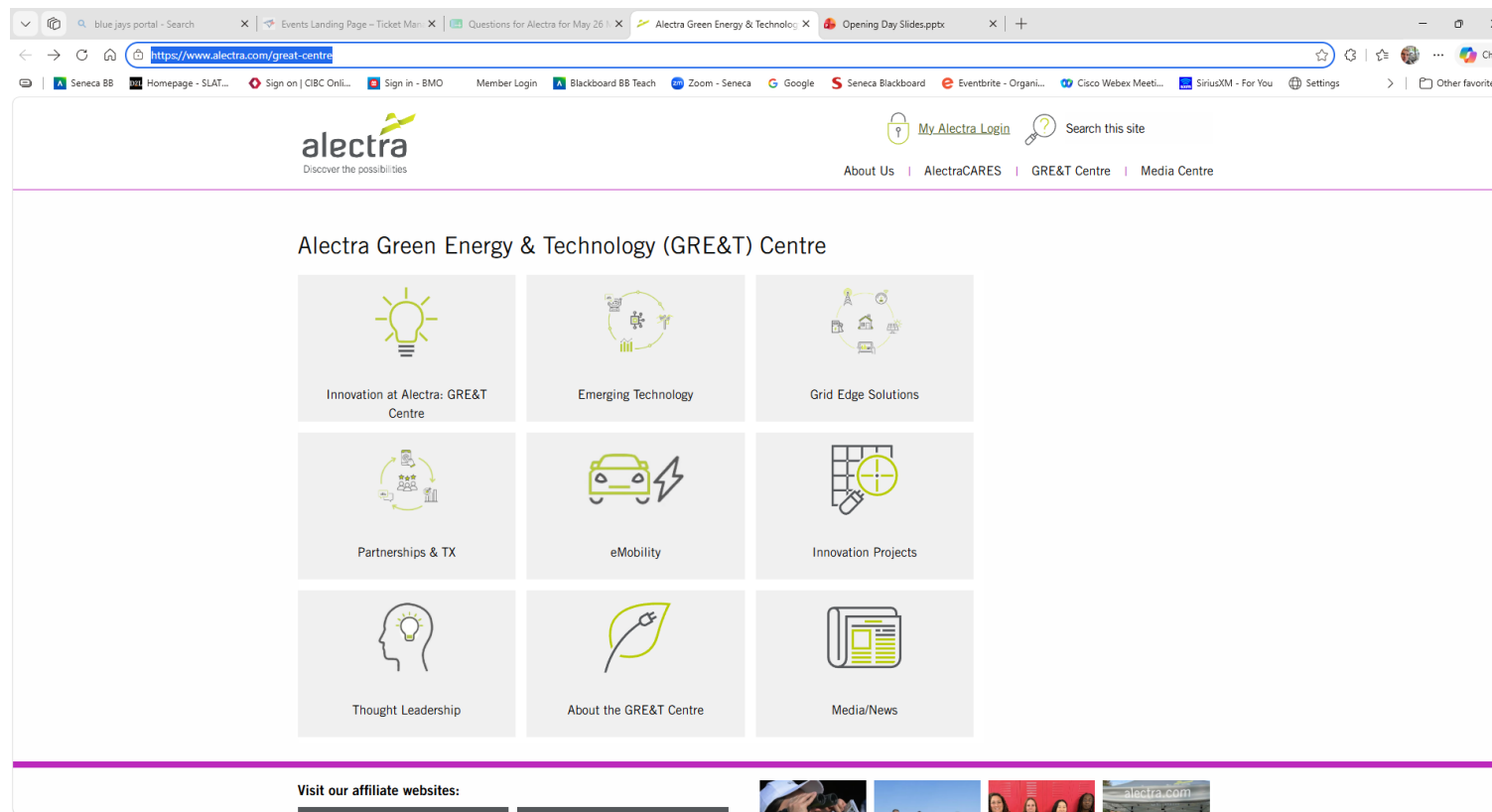
*Design Thinking + GIS + Data + Creativity = Meaningful Innovation*

# SUGGESTION

## RESEARCH THE INDUSTRY SECTOR

### Alectra Green Energy & Technology (GRE&T) Centre

*Pre-read recommended :* [Video Link](https://www.alectra.com/innovation-projects) | <https://www.alectra.com/innovation-projects>



## SUGGESTION

### RESEARCH AVAILABLE TECHNOLOGIES

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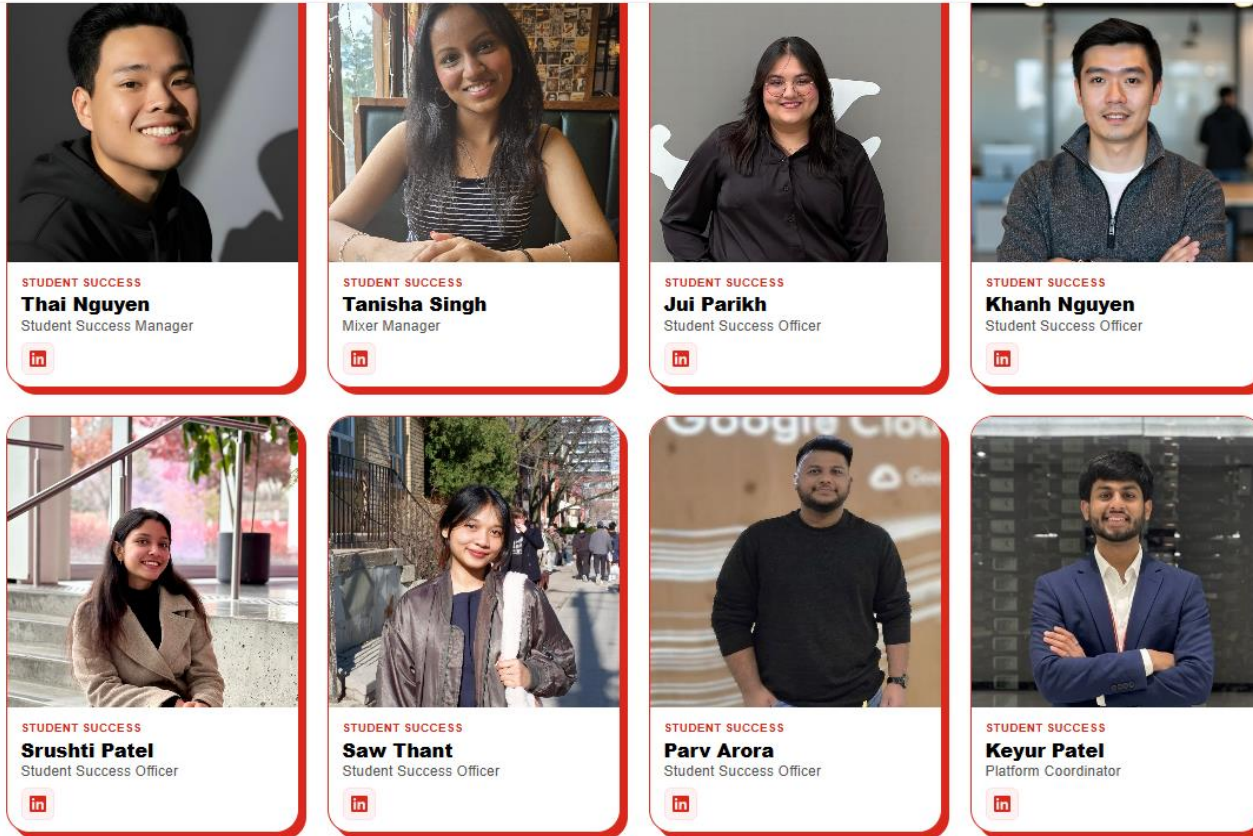
- Liberal use of AI tools such as Copilot, ChatGPT and Claude
- Google and official websites
- YouTube videos and tutorials
- Apply studies from school — Systems Analysis & Design, Project Management, Data Analytics, Programming, Web Development, Software Engineering, UX/UI, Engineering, Business

**ATTEND MENTORSHIP SESSIONS**

**GIVE FEEDBACK TO YOUR  
STUDENT SUCCESS OFFICER**

PLEASE

## LEVERAGE THE STUDENT SUCCESS TEAM



Your Student Success Team is here to support you throughout the hackathon. Reach out through any of these channels:

- Discord
- Email
- Text
- WhatsApp
- Phone calls

# RECOGNITION

## MORE THAN PRIZES

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- **Recognition & Prestige** — Top 12 Finalist Recognition, Judging Panel Exposure, Sponsor Visibility, LinkedIn Recognition, Innovation Awards, Showcase Opportunities
- **Resume & Portfolio Building** — Real-world project experience, professional presentations, GIS & Design Thinking artifacts, dashboards & prototypes, UML diagrams, video pitch portfolio pieces
- **Career & Networking Opportunities** — Industry mentor connections, sponsor interaction, utility & technology exposure, faculty recommendations, applied research visibility, professional networking with Esri Canada and Alectra
- **Learning & Skill Development** — GIS experience, design thinking, data storytelling, team collaboration, systems analysis & design, presentation & leadership skills
- **Pride & Personal Growth** — Challenge yourself, build confidence, create something meaningful, work under pressure, lead a team, deliver impact
- **Real-World Impact** — Energy sustainability, climate resilience, community equity, smart infrastructure, Ontario energy innovation, solutions that matter

*The Real Prize: Experience + Recognition + Skills + Impact + Community*

## NEXT STEPS

### WHAT TO DO

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1

#### **Pick a Challenge Statement**

Choose from the three themes below

2

#### **Research & Develop Your Proposal**

Dig into your subject area

3

#### **Communicate With Your Team & Hack!**

Collaborate and build

#### THE THREE THEMES

1. Clean Energy Generation & Integration
2. Smart Grid Resilience & Electrification
3. Community Energy, Equity & Sustainability



# SUBMISSION

## YOUR DELIVERABLE

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### A five-minute YouTube video and supporting artifacts:

- Design documents
  - Code
  - Wireframes
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- You are encouraged to use AI for research and design.
  - **A submission consisting of AI slop in any form will disqualify your solution.**
  - Do not artificially meet the 5-minute target by speeding up your video.

## QUALIFIER ROUND

### WHAT TO SUBMIT

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**YouTube video (maximum 6 minutes) submitted via the website. Your video should:**

- Define business domains and stakeholders
- Identify real-world impacts
- Keep the story cohesive
- Explain why the idea is feasible and beneficial
- Use diagrams or models to illustrate the system
- Show a realistic and functional demonstration

**DEADLINE: TUESDAY, MARCH 26, 2026, AT 11:59 PM**

# JUDGING

## HOW ARE YOU SCORED

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-  Innovation & Creativity
-  Impact & Relevance
-  Technical Execution
-  Presentation & Communication
-  Collaboration & Teamwork

# GOOD QUALIFIER SUBMISSIONS

## WHAT MAKES A STRONG ROUND 1 SUBMISSION

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### STRONG SUBMISSIONS

- Authentic & Human — speak naturally, look at the camera, show passion and ownership
- Clear Team Roles — briefly introduce members and explain contributions
- Focused Problem Solving — select ONE meaningful use case, narrow scope intelligently
- Demonstrate Something — show mockups, dashboards, GIS maps, wireframes, prototypes
- Reflect & Learn — discuss lessons learned, challenges, and what you would improve

### COMMON MISTAKES

- Reading Slides — robotic delivery, no eye contact, over-scripted
- Repeating the Challenge Set — judges already know the problem
- Trying to Solve Everything — overly broad, no clear use case, unrealistic scope
- Too Much Theory — long explanations with no visuals or demonstration
- Weak Storytelling — no stakeholder focus, no impact explanation, no emotional connection

*Video structure: 0:00 Team Intro → 0:30 Problem & Stakeholder → 1:15 Solution & Use Case → 2:30 Demo/Prototype → 4:00 Impact, Lessons & Future Vision*

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# Q & A SESSION

*Ask us anything*

# CHALLENGE ACCEPTED

**THANK YOU FOR JOINING US**

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Stay connected · [hackathon@senecapolytechnic.ca](mailto:hackathon@senecapolytechnic.ca)